

Fundamentals of Programming:

1. Getting Started
2. Basics
3. Operators
4. Control Flow
5. Objects
6. Arrays
7. Function

Advanced Topics:

1. Gettings Started
2. Objects
3. Prototypes
4. Prototypical Inheritance
5. ES6 Classes
6. ES6 Tooling

Unit Testing:

1. Getting Started
2. Introduction to Unit Testing
3. Core Unit Testing Techniques
4. Breaking Dependencies with Mocks
5. Improving Code Quality with Static Analysis Tools
6. Conclusion
7. What Next?

Topics which You Will Learn Are:

Basics of Programming

- Hello world program
- Declare variable using var
- More about variable, Let, Const
- String Indexing
- Useful string methods, Template Strings
- Null, undefined, BigInt, typeof
- Booleans and Comparison Operator
- Truthy and Falsy Values
- If else statement
- Ternary Operator, && || operator
- Nested if else, If elseif else, Switch statement
- While loop, While loop examples, For loop, For loop examples
- Break and continue keyword
- Do while loop

Arrays

- Intro to arrays
- Push pop shift unshift
- Primitive vs reference data types
- Clone array & spread operator
- For loop
- use const for creating arrays
- While loop in array
- For of loop
- For in loop
- Array destructuring

Objects

- Intro to objects
- Dot vs Bracket Notation
- Iterate objects
- Computed properties
- Spread operator in objects
- Object Destructuring

- Objects inside Array
- Nested Destructuring

Functions

- Function declaration, Expression, Arrow
- Function declarations are hoisted (covered in great detail, later in this course)
- Function inside function
- Lexical Scope
- Block Scope Vs Function Scope
- Default Parameters
- Rest Parameters
- Parameter DE structuring
- Very brief intro to callback functions (covered in great detail, later in this course)
- Functions returning Functions

Very Important Array Methods

- Foreach method
- Map method
- Filter
- Reduce
- Sort
- Find
- Every
- Some
- Fill method
- Splice method

More useful things

- Iterables
- Sets
- Maps
- Object.assign
- Optional chaining

Object Oriented Programming / Prototypal Inheritance

- Methods
- This keyword, Window object
- Call , apply and bind method
- Some warnings
- This inside arrow functions
- Short syntax for methods
- Factory functions & discuss some memory related problems
- First solution to that problem
- Why that solution isn't that great
- What is `__proto__` , `[[prototype]]`
- What is prototype
- Use prototype
- New keyword
- Constructor function with new keyword
- More discussion about proto and prototype
- Class keyword
- Example using class keyword
- Super keyword
- Method overriding
- Getters and setters
- Static methods and properties

How JavaScript Works

- Global Execution context
- This and window in global execution context
- Hoisting
- Are let and const are hoisted ? What is a reference Error ?
- Function execution context
- Scope chain and lexical environment

Closures

- Intro to closures
- Closure example 1
- Closure Example 2
- Closure Example 3

DOM Tutorial

- HTML and CSS Crash course (Around 30-40 minutes)
- Async vs defer
- Select elements using id
- querySelector
- textContent & innerText
- Change the styles of elements using js
- Get and set attributes
- Select multiple elements and loop through them
- innerHTML
- Deeply understand dom tree, root node , element nodes, text nodes
- classList
- Add new elements to page
- Create elements
- Insert adjacent elements
- Clone nodes
- More methods to add elements on page
- How to get the dimensions of the element

Events

- Intro to events
- This keyword inside eventListener callback
- Add events on multiple elements
- Event object
- How event listener works
- Practice with events
- Create demo project
- More events
- Event bubbling
- Event Capturing
- Event delegation
- Create Project using event delegation

Asynchronous

- Is Javascript a synchronous or asynchronous programming language ?

- `setTimeout()`
- `setTimeout()` with 0 millisecond
- Callback Queue
- `setInterval` and create little project with `setInterval`
- Understand callbacks in general
- Callbacks in asynchronous programming
- Callback Hell and Pyramid of dom

Promises, network request

- Intro to promises
- Microtask Queue
- Function that returns promise
- Promise and set timeout
- Promise. Resolve and more about then method
- Convert nested Callbacks to flat code using promises
- Intro to Ajax, HTTP Request
- XHR requests
- Error handling in XHR requests
- XHR request Chaining
- Remodifying XHR requests and chaining using then method
- Fetch API
- Error Handling in Fetch API
- Consume Promises with `async` and `Await`

ES6 (ES2015) Modules

1. Split code into multiple files using ES6 modules.
2. Congratulations
3. Now you know javascript in Great Details
4. What next ?
5. Multiple Projects
6. Interview Questions
7. Web Performance
8. Testing

**Almost no cost
course and with
live classes on
daily bases you
will love it!.**

NaseebCodeStudio